

Explanation:-

When a light ray fall on a smooth shiny surface as a mirror, light ray bounces back into the same medium. Bouncing back of light into the same medium after striking on an Opaque obstacle is known as reflection. But when the ray bounces back, the path of the ray of light gets changed.

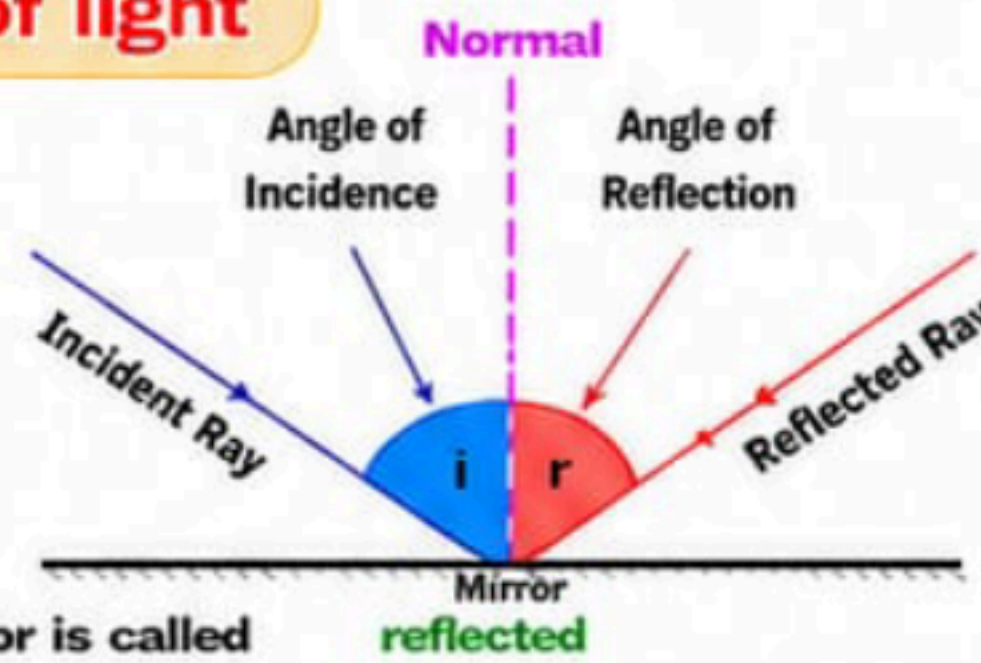
The ray which is strike the surface of the mirror is

called **incident ray** and the ray emerging out after reflection from the mirror is called

When parallel light rays falls on a shiny mirror or still water surface, the reflected rays, also come out as parallel, but when the same parallel rays fall on a dull surface such as a wall, the reflected rays are not parallel but get scattered in all directions.

It is a smooth regular reflection from a shiny is smooth surface that forms an image while the diffused reflection from an irregular and dull surface fails to form an image due to the diffused reflection

Reflection of light



Laws of reflection

First law: Angle of incidence is equal to angle of reflection.

Formula :

$$\text{Angle } i = \text{Angle } r$$

Second law: The incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane.

Formation of images

The picture of our face that we see in a mirror is called an **image**. This images formed when light through our face gets reflected. When you observe carefully you will find that the image of your face appears to be formed behind the mirror. The image formed by reflection of light are of two types:

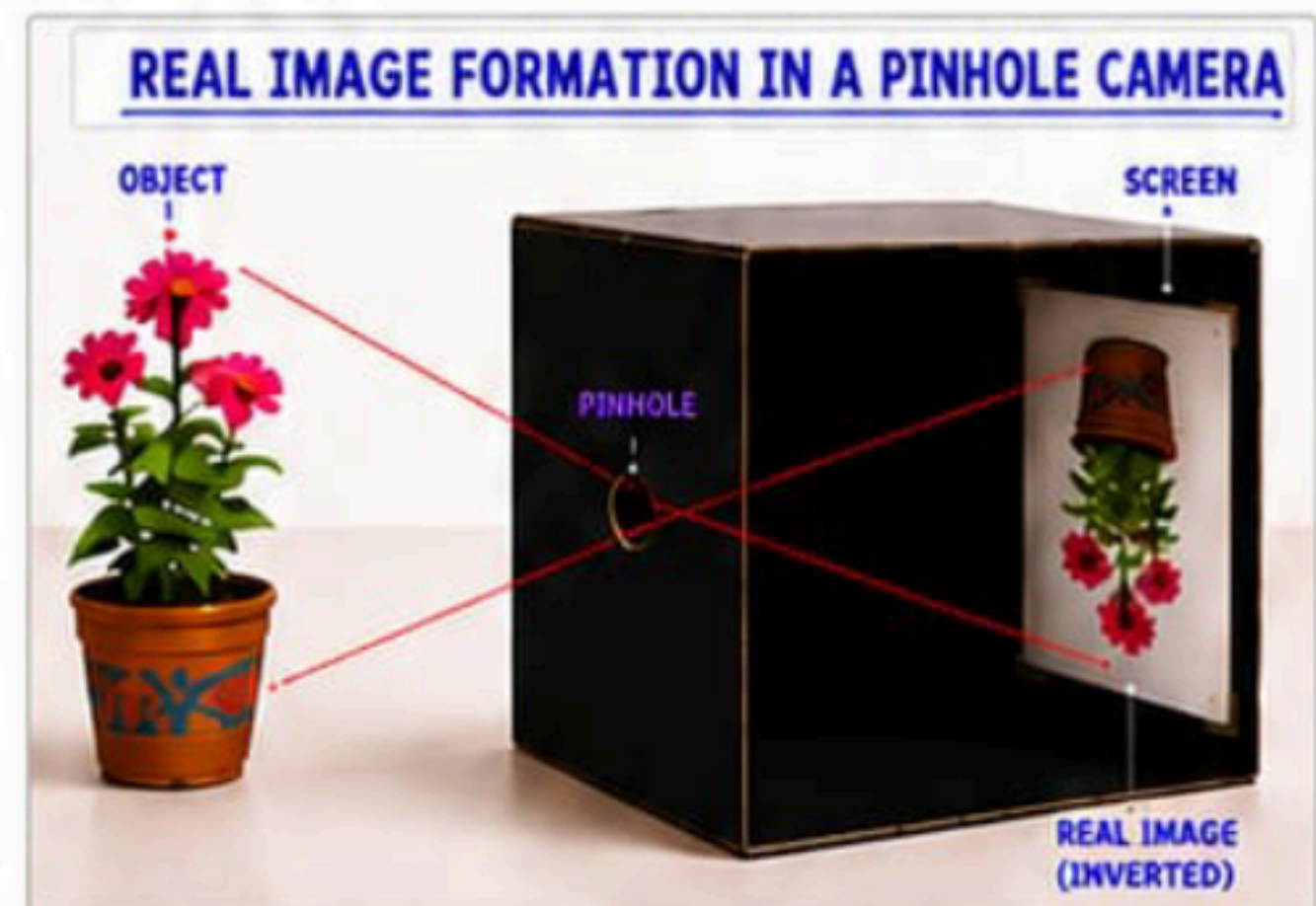
- Real image
- Virtual image

Real images

The image that can be obtained on a screen is called a **real image**.

The image obtained in a pinhole camera is a real image. Also the images obtained on a cinema screen are real images. A real image is found when the rays after reflection actually meet. Generally, curved mirrors form real images. Image formed in the inward surface of a spoon is a real image.

Real image is always **inverted**.



Virtual images

Image that cannot be obtained on a screen is known as **virtual image**. The image formed by plane mirror is a virtual image. In simple words, virtual image is just an illusion. A virtual image is formed when the light rays after reflection appear to meet behind the mirror.

Some important characteristics of the image formed by a plane mirror are:

1. When you stand in front of a mirror and move forward and backward, you will find that the image also moves the same way. Thus, the image is formed at the same distance behind the mirror as the object is in front of it.
2. You might have often noticed that in front of a mirror, your right hand appears to be left and left appears to be right. This is a characteristic property of a mirror that it forms a image that is laterally inverted.

The left - right reversal of an image formed by a mirror is known as **lateral inversion**. This clearly justifies why **AMBULANCE** is written laterally inverted.

The word 'AMBULANCE' in front of is written in the reverse order so that when the driver in the front vehicle looks in his rear view mirror which is a plane mirror, they can see the laterally inverted image of ambulance and can read it correctly. Then they can immediately give way to the ambulance.

This phenomenon of left-right reversal in a plane mirror is called **lateral inversion**.



3. We see our image to be of the same size as we are. Thus, the image formed is of the same size as that of the object.
4. When we see ourselves in the mirror we see our image to be just the way we are, and not inverted. Thus, the image formed by a plane mirror is **upright or erect**.

Thus, all the characteristics of the image formed by a plane mirror can be written as:

- Images virtual and erect.
- Image is of the same size as that of the object.
- Images formed as much behind the mirror as the object is in front of it.
- Image formed is laterally inverted.
- Left - right reversal of image (lateral inversion).

Uses of a plane mirrors

- We use plane mirrors as looking glass.
- Mirrors are also used for decoration.
- Mirrors are used to increase the effective length in showrooms.
- Periscope also use plane mirror.

